Electromagnetic Waves

Single Correct Type Questions

1. If $\vec{E}$ and $\vec{K}$ represent electric field and propagation vectors of the EM waves in vacuum, then magnetic field vector is given by: (ω – angular frequency):

[24 Jan, 2023 (Shift-I)]

(*a*) $\frac{1}{\omega}\left(\vec{K}\times\vec{E}\right)$ (*b*) $\omega\left(\vec{E}\times\vec{K}\right)$

(*c*) $\omega\left(\vec{K}\times\vec{E}\right)$ (*d*) $\vec{K}\times\vec{E}$

2. An EM wave propagating in x-direction has a wavelength of 8 mm. The electric field vibrating y-direction has maximum magnitude of 60 Vm^{-1}. Choose the correct equations for electric and magnetic field if the EM wave if propagating in vacuum **[28 June, 2022 (Shift-II)]**

(*a*) $E_y = 60\sin\left[\frac{\pi}{4}\times10^3(x-3\times10^8 t)\right]\hat{j}\,Vm^{-1}$

$B_z = 2\sin\left[\frac{\pi}{4}\times10^3(x-3\times10^8 t)\right]\hat{k}T$

(*b*) $E_y = 60\sin\left[\frac{\pi}{4}\times10^3(x-3\times10^8 t)\right]\hat{j}\,Vm^{-1}$

$B_z = 2\times10^{-7}\sin\left[\frac{\pi}{4}\times10^3(x-3\times10^8 t)\right]\hat{k}T$

(*c*) $E_y = 2\times10^{-7}\sin\left[\frac{\pi}{4}\times10^3(x-3\times10^8 t)\right]\hat{j}\,Vm^{-1}$

$B_z = 60\sin\left[\frac{\pi}{4}\times10^3(x-3\times10^8 t)\right]\hat{k}T$

(*d*) $E_y = 2\times10^{-7}\sin\left[\frac{\pi}{4}\times10^4(x-4\times10^8 t)\right]j\,Vm^{-1}$

$B_z = 60\sin\left[\frac{\pi}{4}\times10^4(x-4\times10^8 t)\right]\hat{k}T$

3. If Electric field intensity of a uniform plane electro magnetic wave is given as

$$E = -301.6\sin(kz-\omega t)\hat{a}_x + 452.4\sin(kz-\omega t)\hat{a}_y\,\frac{V}{m}$$

Then, magnetic intensity H of this wave in Am^{-1} will be:
[Given: Speed of light in vaccum $\mu_0 = 4\pi\times10^{-7}$ NA^{-2}, Permeability of vacuum] **[26 June, 2022 (Shift-I)]**

(*a*) $+0.8\sin(kz-\omega t)\hat{a}_y + 0.8\sin(kz-\omega t)\hat{a}_x$

(*b*) $+1.0\times10^{-6}\sin(kz-\omega t)\hat{a}_y + 1.5\times10^{-6}(kz-\omega t)\hat{a}_x$

(*c*) $-0.8\sin(kz-\omega t)\hat{a}_y - 1.2\sin(kz-\omega t)\hat{a}_x$

(*d*) $-1.0\times10^{-6}\sin(kz-\omega t)\hat{a}_y - 1.5\times10^{-6}\sin(kz-\omega t)\hat{a}_x$

4. Electric field in a plane electromagnetic wave is given by E = 50 sin $(500x - 10\times10^{10}t)$V/m

[27 Aug, 2021 (Shift-I)]

The velocity of electromagnetic wave in this medium is :
(Given c = speed of light in vacuum)

(*a*) c (*b*) $\frac{c}{2}$ (*c*) $\frac{2}{3}c$ (*d*) $\frac{3}{2}c$

5. A plane electromagnetic wave is propagating along the direction $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$, with its polarization along the direction $\hat{k}$. The correct form of the magnetic field of wave would be (here B_0 is an appropriate constant)

[9 Jan, 2020 (Shift-II)]

(*a*) $B_0\frac{\hat{i}+\hat{j}}{\sqrt{2}}\cos\left(\omega t - k\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

(*b*) $B_0\frac{\hat{i}-\hat{j}}{\sqrt{2}}\cos\left(\omega t + k\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

(*c*) $B_0\frac{\hat{i}-\hat{j}}{\sqrt{2}}\cos\left(\omega t - k\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

(*d*) $B_0\hat{k}\cos\left(\omega t - k\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

6. Which of the following Maxwell's equations is valid for time varying conditions but not valid for static conditions: **[13 Apr 2023 (Shift-I)]**

(a) $\oint \vec{B}\cdot\vec{dl} = \mu_0 I$ (b) $\oint \vec{E}\cdot\vec{dl} = 0$

(c) $\oint \vec{E}\cdot\vec{dl} = -\frac{\partial\phi_B}{\partial t}$ (d) $\oint \vec{D}\cdot\vec{dA} = Q$

7. Match List-I with List-II: **[25 Jan 2023 (Shift-II)]**

A.	Gauss's Law in Electrostatics	I.	$\oint \vec{E}.\vec{dl} = -\frac{d\phi_B}{dt}$
B.	Faraday's Law	II.	$\oint \vec{B}\cdot d\vec{A} = 0$
C.	Gauss's Law in Magnetism	III.	$\oint \vec{B}\cdot\vec{l} = \mu_0 i_C + \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$
D.	Ampere Maxwell Law	IV.	$\oint \vec{E}\cdot\vec{s} = \frac{q}{\epsilon_0}$

Choose the correct answer from the options given below:

(a) A-IV, B-I, C-II, D-III (b) A-I, B-II, C-III, D-IV

(c) A-III, B-IV, C-I, D-II (d) A-II, B-III, C-IV, D-I

8. The electric field and magnetic field components of an electromagnetic wave going through vacuum is described by **[24 Jan, 2023 (Shift-II)]**

$$E_x = E_0 \sin(kz - \omega t)$$

$$B_y = B_0 \sin(kz - \omega t)$$

Then the correct relation between E_0 and B_0 is given by

(a) $kE_0 = \omega B_0$ (b) $E_0 B_0 = \omega k$

(c) $\omega E_0 = kB_0$ (d) $E_0 = kB_0$

9. Given below are two statements:

Statement-I: Electromagnetic waves are not deflected by electric and magnetic field.

Statement-II: The amplitude of electric field and the magnetic field in electromagnetic waves are related to each other as $E_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} B_0$

In the light of the above statements, choose the correct answer from the options given below:

[29 Jan, 2023 (Shift-II)]

(a) Statement-I is true but Statement-II is false

(b) Both Statement-I and Statement-II are true

(c) Statement-I is false but Statement-II is true

(d) Both Statement-I and Statement-II are false

10. The electric field associated with an electromagnetic wave propagating in a dielectric medium is given by $\vec{E} = 30(2\hat{x} + \hat{y})\sin\left[2\pi\left(5\times10^{14}t - \frac{10^7}{3}z\right)\right] Vm^{-1}$ Which of the following option(s) is(are) correct?

[Given: The speed of light in vacuum, $c = 3 \times 10^8$ ms^{-1}]

[JEE Adv, 2023]

(a) $B_x = -2\times10^{-7}\sin\left[2\pi\left(5\times10^{14}t - \frac{10^7}{3}z\right)\right]$ Wbm^{-2}

(b) $B_y = 2\times10^{-7}\sin\left[2\pi\left(5\times10^{14}t - \frac{10^7}{3}z\right)\right]$ Wbm^{-2}

(c) The wave is polarized in the xy-plane with polarization angle 30° with respect to the x-axis.

(d) The refractive index of the medium is 4.

11. The magnetic field of a plane electromagnetic wave is given by: **[26 July, 2022 (Shift-I)]**

$$\vec{B} = 2\times10^{-8}\sin(0.5\times10^3 x + 1.5\times10^{11} t)\hat{j}T$$

The amplitude of the electric field would be:

(a) 6Vm^{-1} along x-axis

(b) 3Vm^{-1} along z-axis

(c) 6Vm^{-1} along z-axis

(d) 2 $\times 10^{-8}$ Vm^{-1} along z-axis

12. The oscillating magnetic field in a plane electromagnetic wave is given by $B_y = 5 \times 10^{-6} \sin 100\pi\,(5x - 4 \times 10^8\, t)T$. The amplitude of electric field will be:

[26 July, 2022 (Shift-II)]

(a) 15 $\times 10^2$ Vm^{-1}

(b) 5 $\times 10^{-6}$ Vm^{-1}

(c) 16 $\times 10^{12}$ Vm^{-1}

(d) 4 $\times 10^2$ Vm^{-1}

13. Light wave traveling in air along x-direction is given by $E_y = 540 \sin\pi\times10^4(x - ct)$Vm^{-1}. Then, the peak value of magnetic field of wave will be: (Given, $c = 3 \times 10^8$ ms^{-1})

[25 July, 2022 (Shift-II)]

(a) 18 $\times 10^{-7}$ T

(b) 54 $\times 10^{-7}$ T

(c) 54 $\times 10^{-8}$ T

(d) 18 $\times 10^{-8}$ T

14. The electric field of a plane electromagnetic wave propagating along the x direction in vaccum is $\vec{E} = E_0 \cos(\omega t - kx)\hat{j}$. The magnetic field $\vec{B}$, at the moment $t = 0$ is **[3 Sep, 2020 (Shift-II)]**

(a) $\vec{B} = E_0\sqrt{\mu_0\varepsilon_0}\cos(kx)\hat{j}$

(b) $\vec{B} = E_0\sqrt{\mu_0\varepsilon_0}\cos(kx)\hat{k}$

(c) $\vec{B} = \dfrac{E_0}{\sqrt{\mu_0\varepsilon_0}}\cos(kx)\hat{k}$

(d) $\vec{B} = \dfrac{E_0}{\sqrt{\mu_0\varepsilon_0}}\cos(kx)\hat{j}$

15. For a plane electromagnetic wave, the magnetic field at a point x and time t is **[6 Sep, 2020 (Shift-II)]**

$\vec{B}(x,t) = [1.2\times10^{-7}\sin(0.5\times10^3 x + 1.5\times10^{11}t)\hat{k}]T.$

The instantaneous electric field $\vec{E}$ corresponding to $\vec{B}$ is (speed of light $c = 3 \times 10^8$ ms^{-1})

(a) $\vec{E}(x,t) = [36\sin(1\times10^3 x + 1.5\times10^{11}t)\hat{i}]\dfrac{V}{m}$

(b) $\vec{E}(x,t) = [36\sin(0.5\times10^3 x + 1.5\times10^{11}t)\hat{k}]\dfrac{V}{m}$

(c) $\vec{E}(x,t) = [-36\sin(0.5\times10^3 x + 1.5\times10^{11}t)\hat{j}]\dfrac{V}{m}$

(d) $\vec{E}(x,t) = [36\sin(1\times10^3 x + 1.5\times10^{11}t)\hat{j}]\dfrac{V}{m}$

16. The magnetic field of a plane electromagnetic wave is

$\vec{B} = 3 \times 10^{-8}\sin[200\,\pi(y + ct)]\,\hat{i}T$

Where $c = 3 \times 10^8$ ms^{-1} is the speed of light.

The corresponding electric field is

[03 Sep, 2020 (Shift-I)]

(a) $\vec{E} = 9\sin\left[200\pi\left(y + ct\right)\right]\hat{k}V/m$

(b) $\vec{E} = -9\sin\left[200\pi\left(y + ct\right)\right]\hat{k}V/m$

(c) $\vec{E} = 3\times10^{-8}\sin\left[200\pi\left(y + ct\right)\right]\hat{k}V/m$

(d) $\vec{E} = -10^{-6}\sin\left[200\pi\left(y + ct\right)\right]\hat{k}V/m$

17. If the magnetic field in a plane electromagnetic wave is given by $\vec{B} = 3 \times 10^{-8}\sin(1.6 \times 10^3 x + 48 \times 10^{10}t)\,\hat{j}$ T, then what will be expression for electric field? **[7 Jan, 2020 (Shift-I)]**

(a) $\vec{E} = (3 \times 10^{-8}\sin(1.6 \times 10^3 x + 48 \times 10^{10}t)\,\hat{j}$ V/m)

(b) $\vec{E} = (3 \times 10^{-8}\sin(1.6 \times 10^3 x + 48 \times 10^{10}t)\,\hat{i}$ V/m)

(c) $\vec{E} = (60\sin(1.6 \times 10^3 x + 48 \times 10^{10}t)\,\hat{k}$ V/m)

(d) $\vec{E} = (9\sin(1.6 \times 10^3 x + 48 \times 10^{10}t)\,\hat{k}$ V/m)

18. The electric fields of two plane electromagnetic plane waves in vacuum are given by $\vec{E}_1 = E_0\hat{j}\cos(\omega t - kx)$ and $\vec{E}_2 = E_0\hat{k}\cos(\omega t - ky)$. At $t = 0$, a particle of charge q is at origin with a velocity $\vec{v} = (0.8c)\hat{j}$ (c is the speed of the light in vacuum). The instantaneous force experienced by the particle is **[9 Jan, 2020 (Shift-I)]**

(a) $E_0q(0.8\hat{i} + \hat{j} + 0.2\hat{k})$

(b) $E_0q(-0.8\hat{i} + \hat{j} + \hat{k})$

(c) $E_0q(0.4\hat{i} - 3\hat{j} + 0.8\hat{k})$

(d) $E_0q(0.8\hat{i} - \hat{j} + 0.4\hat{k})$

19. The electric field of a plane electromagnetic wave is given by $\vec{E} = E_0\hat{i}\cos(kz)\cos(\omega t)$. The corresponding magnetic field $\vec{B}$ is then given by **[10 April, 2019 (Shift-I)]**

(a) $\vec{B} = \dfrac{E_0}{c}\hat{j}\sin(kz)\cos(\omega t)$

(b) $\vec{B} = \dfrac{E_0}{c}\hat{j}\sin(kz)\sin(\omega t)$

(c) $\vec{B} = \dfrac{E_0}{c}\hat{k}\sin(kz)\cos(\omega t)$

(d) $\vec{B} = \dfrac{E_0}{c}\hat{j}\cos(kz)\sin(\omega t)$

20. The magnetic field of an electromagnetic wave is given by

$\vec{B} = 1.6\times10^{-6}\cos\left(2\times10^7 z + 6\times10^{15}t\right)\left(2\hat{i} + \hat{j}\right)\dfrac{Wb}{m^2}$

The associated electric field will be

[8 April, 2019 (Shift-II)]

(a) $\vec{E} = 4.8\times10^2\cos\left(2\times10^7 z + 6\times10^{15}t\right)\left(\hat{i} - 2\hat{j}\right)\dfrac{V}{m}$

(b) $\vec{E} = 4.8\times10^2\cos\left(2\times10^7 z - 6\times10^{15}t\right)\left(2\hat{i} + \hat{j}\right)\dfrac{V}{m}$

(c) $\vec{E} = 4.8\times10^2\cos\left(2\times10^7 z - 6\times10^{15}t\right)\left(-2\hat{j} + \hat{i}\right)\dfrac{V}{m}$

(d) $\vec{E} = 4.8\times10^2\cos\left(2\times10^7 z + 6\times10^{15}t\right)\left(-\hat{i} + 2\hat{j}\right)\dfrac{V}{m}$

21. The electric field of a plane polarized electromagnetic wave in free space at time $t = 0$ is given by an expression

$\vec{E}(x, y) = 10\hat{j}\cos[6x + 8z]$

The magnetic field $B(x, z, t)$ is given by (c is the velocity of light) **[10 Jan, 2019 (Shift-II)]**

(*a*) $\frac{1}{c}(6\hat{k} + 8\hat{i})\cos[(6x - 8z + 10ct)]$

(*b*) $\frac{1}{c}(6\hat{k} - 8\hat{i})\cos[(6x + 8z - 10ct)]$

(*c*) $\frac{1}{c}(6\hat{k} + 8\hat{i})\cos[(6x + 8z - 10ct)]$

(*d*) $\frac{1}{c}(6\hat{k} - 8\hat{i})\cos[(6x + 8z + 10ct)]$

22. An electromagnetic wave of intensity 50 Wm^{-2} enters in a medium of refractive index 'n' without any loss. The ratio of the magnitudes of electric fields, and the ratio of the magnitudes of magnetic fields of the wave before and after entering into the medium are respectively, given by **[11 Jan, 2019 (Shift-I)]**

(*a*) $\left(\frac{1}{\sqrt{n}}, \frac{1}{\sqrt{n}}\right)$ (*b*) $(\sqrt{n}, \sqrt{n})$

(*c*) $\left(\sqrt{n}, \frac{1}{\sqrt{n}}\right)$ (*d*) $\left(\frac{1}{\sqrt{n}}, \sqrt{n}\right)$

23. If the magnetic field of a plane electromagnetic wave is given by (The speed of light = 3×10^8 m/s)

$B = 100 \times 10^{-6} \sin\left[2\pi \times 2 \times 10^{15}\left(t - \frac{x}{c}\right)\right]$ then the maximum electric field associated with it is **[10 Jan, 2019 (Shift-I)]**

(*a*) 6×10^4 N/C
(*b*) 3×10^4 N/C
(*c*) 4×10^4 N/C
(*d*) 4.5×10^4 N/C

24. For the plane electromagnetic wave given by $E = E_0 \sin(\omega t - kx)$ and $B = B_0 \sin(\omega t - kx)$, the ratio of average electric energy density to average magnetic energy density is **[06 April, 2023 (Shift-I)]**

(*a*) 1 (*b*) 1/2
(*c*) 2 (*d*) 4

25. The electric field in an electromagnetic wave is given as

$\vec{E} = 20\sin\omega\left(t - \frac{x}{c}\right)\vec{j}NC^{-1}$

Where ω and C are angular frequency and velocity of electromagnetic wave respectively. The energy contained in a volume of 5×10^{-4} m^3 will be
(Given $\varepsilon_0 = 8.85 \times 10^{-12}$ C^2 / Nm^2) **[11 April, 2023 (Shift-I)]**

(*a*) 28.5×10^{-13}J (*b*) 17.7×10^{-13}J
(*c*) 8.85×10^{-13}J (*d*) 88.5×10^{-13}J

26. A plane electromagnetic wave, has frequency of 2.0×10^{10} Hz and its energy density is 1.02×10^{-8} J/m^3 in vacuum. The amplitude of the magnetic field of the wave is close to

($\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9 \frac{Nm^2}{C^2}$ and speed of light = 3×10^8 ms^{-1}) **[2 Sep, 2020 (Shift-I)]**

(*a*) 110 nT (*b*) 140 nT
(*c*) 190 nT (*d*) 160 nT

27. The dimension of $\frac{B^2}{2\mu_0}$, where B is magnetic field and μ_0 is the magnetic permeability of vacuum, is **[7 Jan, 2020 (Shift-II)]**

(*a*) $ML^2 T^{-1}$ (*b*) $ML^{-1} T^{-2}$
(*c*) $ML^2 T^{-1}$ (*d*) $ML T^{-2}$

28. Match the List-I with List-II: **[01 Feb 2023 (Shift-I)]**

	List-I		List-II
A.	Microwaves	I.	Radio active decay of the nucleus
B.	Gamma rays	II.	Rapid acceleration and deceleration of electron in aerials
C.	Radio waves	III.	Inner shell electrons
D.	X-rays	IV.	Klystron valve

Choose the correct answer from the options given below:
(*a*) A → I, B → II, C → III, D → IV
(*b*) A → IV, B → I, C → II, D → III
(*c*) A → I, B → III, C → IV, D → II
(*d*) A → IV, B → III, C → II, D → I

29. Match Column-I and Column-II

	Column-I		Column-II
A.	Ultraviolet rays	p.	Study crystal
B.	Microwaves	q.	Greenhouse effect
C.	Infrared waves	r.	Sterilizing surgical instrument
D.	X-rays	s.	Radar system

Choose the correct answer from the option given below **[27 June, 2022 (Shift-I)]**

(*a*) A-(r); B-(s); C-(q); D-(p)
(*b*) A-(q); B-(p); C-(s); D-(r)
(*c*) A-(r); B-(q); C-(p); D-(s)
(*d*) A-(q); B-(p); C-(r); D-(s)

Integer Type Question

30. The peak electric field produced by the radiation coming from the 8 W bulb at a distance of 10 m is $\frac{x}{10}\sqrt{\frac{\mu_0 c}{\pi}\frac{V}{m}}$.

The efficiency of the bulb is 10% and it is a point source. The value of x is……. **[25 Feb, 2021 (Shift-II)]**

ANSWER KEY

1. (*a*)	**2.** (*b*)	**3.** (*c*)	**4.** (*c*)	**5.** (*c*)	**6.** (*c*)	**7.** (*a*)	**8.** (*a*)	**9.** (*a*)	**10.** (*a*)
11. (*c*)	**12.** (*d*)	**13.** (*a*)	**14.** (*b*)	**15.** (*c*)	**16.** (*b*)	**17.** (*d*)	**18.** (*a*)	**19.** (*b*)	**20.** (*d*)
21. (*d*)	**22.** (*c*)	**23.** (*b*)	**24.** (*a*)	**25.** (*c*)	**26.** (*d*)	**27.** (*b*)	**28.** (*b*)	**29.** (*a*)	**30.** [2]

EXPLANATIONS

1. (*a*) Direction of magnetic field vector will be in the direction of $\hat{K}\times\hat{E}$

magnitude of magnetic field $B=\frac{E}{C}=\frac{K}{\omega}E$

$\Rightarrow \vec{B}=\frac{1}{\omega}\left(\vec{K}\times\vec{E}\right)$

2. (*b*) $\hat{E}\times\hat{B}$ must be direction of propagation.

3. (*c*) $\vec{H}=\frac{\vec{B}}{\mu_0}\Rightarrow \vec{B}=\frac{1}{c}\left(\vec{k}\times\vec{E}\right)$

4. (*c*) On comparing with equation, $E = E_0 \sin(kx - \omega t)$, we get

$\omega = 10\times10^{10}$ rad/s

$k = 500$

$\therefore$ Velocity, $v=\frac{\text{coefficient of time}}{\text{coefficiont of } x}=\frac{\omega}{k}$

$\Rightarrow v=\frac{10\times10^{10}}{500}=2\times10^{8}=\frac{2\times c}{3}$

5. (*c*) EM wave is in direction $\rightarrow \frac{\hat{i}+\hat{j}}{\sqrt{2}}$

Electric field is in direction $\rightarrow \hat{k}$

$\vec{E}\times\vec{B}\rightarrow$ direction of propagation of EM wave

6. (*c*) According to Maxwell's eqn.

$\oint E.dl=-\frac{\partial\phi_B}{\partial t}$

7. (*a*) Gauss's Law of electrostatic is,

$\oint\vec{E}\cdot d\vec{s}=\frac{q}{\epsilon_0}$

Faraday's law of *EMI*, $\oint\vec{E}\cdot d\vec{l}=\frac{-d\phi_B}{dt}$

Gauss's law of magnetism is, $\oint\vec{B}\cdot d\vec{A}=0$

Ampere Maxwell law

$\oint\vec{B}\cdot d\vec{l}=\mu_0 i_C+\mu_0\epsilon_0\frac{d\phi_E}{dt}$

8. (*a*) $\because C=\frac{\omega}{k}$

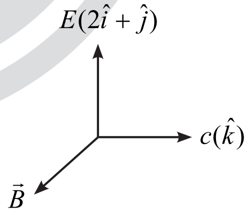
$\Rightarrow \frac{\omega}{k}=\frac{E_0}{B_0}$

9. (*a*) EM waves are neutral and $E_0=\sqrt{\frac{1}{\mu_0\varepsilon_0}}B_0$

Statement-I is correct and

Statement-II is wrong.

10. (*a*)



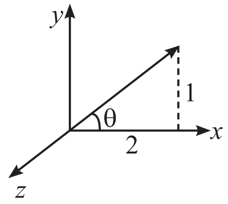
$c_{\text{med.}}=\frac{5\times10^{14}}{10^7/3}=1.5\times10^{8}\,m/s$

Reflective index $\mu=\frac{c}{c_{\text{med.}}}=\frac{3\times10^8}{1.50\times10^8}=2$

$c_{\text{med.}}=\frac{|E|}{|B|}\Rightarrow|B|=\frac{|E|}{c_{\text{med.}}}=\frac{30\sqrt{5}}{1.5\times10^{8}}=2\sqrt{5}\times10^{-7}$

$\hat{B}_{\text{direction}}\equiv\hat{k}\times(2\hat{i}+\hat{j})\equiv\frac{2\hat{j}-\hat{i}}{\sqrt{5}}$

$$\therefore \vec{B} = 2\times10^{-7}(-\hat{i}+2\hat{j})\sin\left[27\left(5\times10^{17}t-\frac{10^7}{3}z\right)\right]$$



$\tan\theta = \frac{1}{2}$

11. (*c*) $E_0 = B_0\, c = 2\times10^{-8}\times3\times10^8 = 6\text{V/m}$

Since, $B \to \hat{j}$ and $v \to \hat{i}$

Therefore, $E \to \hat{k}$

Direction will be along z-axis

12. (*d*) We have, $B_y = 5\times10^{-6}\sin100\pi\,(5x - 4\times10^8 t)T$. On comparing with standard equation, $B = B_0\sin(\omega t + kx)$, we get

$B_0 = 5\times10^{-6}$

Speed of wave, $v = \frac{4\times10^8}{5} = 8\times10^7$ $\quad\left[\because v = \frac{\omega}{K}\right]$

$E_0 = V\times B_0 = 8\times10^7\times5\times10^{-6} = 4\times10^2\ \text{Vm}^{-1}$

13. (*a*) Given, $E_y = 540\sin\pi\times10^4\,(x - ct)\ \text{Vm}^{-1}$

$\therefore E_0 = 540\ \text{Vm}^{-1}$

$$c = \frac{E_0}{B_0} \Rightarrow B_0 = \frac{E_0}{c} = \frac{540}{3\times10^8} = 18\times10^{-7}\ \text{T}$$

14. (*b*) $\vec{E}\times\vec{B}\parallel\vec{C}$

Hence $\vec{B}$ should be in $\hat{k}$ direction.

Also, $E_0 = B_0 C$, $C = \frac{1}{\sqrt{\mu_0\varepsilon_0}}$

$$\Rightarrow \vec{B} = E_0\sqrt{\mu_0\varepsilon_0}\cos(kx)\hat{k}$$

15. (*c*) $\vec{B} = 1.2\times10^{-7}\sin(0.5\times10^3 x + 1.5\times10^{11}t)\,\hat{k}\,]\ T.$

Wave is travelling along $-x$ axis and $\vec{B}$ is along $+z$ axis.

$$E_0 = cB_0 = 36\frac{V}{m}$$

$\vec{E}$ must be along $-y$ axis

16. (*b*) $E_0 = B_0 C = 3\times10^8\times3\times10^{-8} = 9\ \text{V/m}$

Pointing vector $\vec{s} = \frac{\vec{E}\times\vec{B}}{\mu_0}$

$\therefore E = -9\sin[200\,\pi(y + ct)]\ \hat{k}$

(along –ve z-axis)

17. (*d*) $\frac{E_0}{B_0} = c$ (speed of light in vacuum)

$E_0 = B_0 c = 3\times10^{-8}\times3\times10^8 = 9\ \text{N/C}$

So $\vec{E} = 9\sin(1.6\times10^3 x + 48\times10^{10}t)\,\hat{k}$ v/m

18. (*a*) Magnetic field vectors associated with this electromagnetic wave are given by

$$\vec{B}_1 = \frac{E_0}{c}\hat{k}a\cos(kx-\omega t)\ \&\ B_2 = \frac{E_0}{c}i\cos(ky-\omega t)$$

$$\vec{F} = q\vec{E} + q(\vec{V}\times\vec{B})$$

$$= q(\vec{E}_1+\vec{E}_2) + q(\vec{V}\times(\vec{B}_1+\vec{B}_2))$$

by putting the value of $\vec{E}_1, \vec{E}_2, \vec{B}_1\ \&\ \vec{B}_2$

The net Lorentz force on the charged particle is

$$\vec{F} = qE_0[0.8\cos(kx-\omega t)\hat{i} + \cos(kx-\omega t)\hat{j} + 0.2\cos(kx-\omega t)\hat{k}]$$

at $t = 0$ and at $x = y = 0$

$$\vec{F} = qE_0[0.8\hat{i} + \hat{j} + 0.2\hat{k}]$$

19. (*b*) $\because\ \vec{E}\times\vec{B}\parallel\vec{v}$

Given that wave is propagating along positive z-axis and $\vec{E}$ along positive x-axis. Hence $\vec{B}$ long y-axis.

From Maxwell equation

$$\frac{\partial E}{\partial z} = -\frac{\partial B}{dt} \text{ and } B_0 = \frac{E_0}{c}$$

Option (*b*) is correct

20. (*d*) $E = cB$

$E = 3\times10^8\times1.6\times10^{-6}\times\sqrt{5}$

$E = 4.8\times10^2\sqrt{5}$

Direction of electric field is along $\left(-\hat{i}+2\hat{j}\right)$

$$\therefore \vec{E} = 4.8\times10^2\cos\left(2\times10^7 z + 6\times10^{15}t\right)\left(-\hat{i}+2\hat{j}\right)\frac{V}{m}$$

21. (*d*) $\vec{E} = 10\hat{j}\cos\left(6x + 8z - 10ct\right)$

$$B_o = \frac{E_o}{c} = \frac{10}{c}$$

$\omega = 10$ c

$\because\ \hat{E}\times\hat{B} = \hat{C}$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & 0 \\ B_x & B_y & B_z \end{vmatrix} = \frac{6\hat{i}+8\hat{j}}{10}$$

$$\Rightarrow B_z\hat{i} - 0\hat{j} - B_x\hat{k} = \frac{3}{5}\hat{i} + \frac{4}{5}\hat{j}$$

$$\Rightarrow B_z = \frac{3}{5}, B_y = 0, B_z = \frac{4}{5}$$

$$\therefore \vec{B} = \frac{1}{C}\left(-8\hat{i} + 6\hat{k}\right)\cos(6x + 8z + 10ct)$$

22. (*c*) $\frac{E_i}{B_i} = c$ ...(i)

$$\frac{E_f}{B_f} = \frac{c}{n} \quad \text{...(ii)}$$

$$\Rightarrow \frac{E_i B_f}{E_f B_i} = n \Rightarrow \frac{E_i}{E_f} = n\frac{B_i}{B_f}$$

Hence ratio will be $\sqrt{n} : \frac{1}{\sqrt{n}}$

23. (*b*) $E = cB$

24. (a) Electric energy density $= \frac{1}{2} \in_0 E_{rms}^2$

Magnetic energy density $= \frac{B_{rms}^2}{2\mu_0}$

$$\therefore \frac{\text{Electric energy density}}{\text{Magnetic energy density}} = \frac{\frac{1}{2} \in_0 E_{rms}^2}{\left(\frac{B_{rms}^2}{2\mu_0}\right)}$$

$$= \left(\frac{E_{rms}}{B_{rms}}\right)^2 \cdot \mu_0 \in_0 \qquad \left[\because c = \frac{1}{\sqrt{\mu_0 \in_0}}\right]$$

$$\left(\frac{E_0}{\sqrt{2}} \times \frac{\sqrt{2}}{B_0}\right)^2 \cdot c^2 = \frac{c^2}{c^2} = 1 \qquad \left[\because \frac{E_0}{B_0} = C\right]$$

25. (*c*) $\vec{E} = 20\sin\omega\left(t - \frac{x}{C}\right)\hat{j} N/C$

Comparing with $E = E_0 \sin\omega\left(t - \frac{x}{c}\right)\hat{J} N/C$

$E_0 = 20\ N/C$

Average energy density of an em wave

Then Energy stored $= \frac{1}{2} \in_0 E_0^2 \times$ (volume)

$$= \frac{1}{2} \times 8.85 \times 10^{-12} \times (20)^2 \times \left(5 \times 10^{-4}\right) \text{J}$$

8.85×10^{-13}J

26. (*d*) $\frac{B^2}{2\mu_0} = 1.02 \times 10^{-8} \Rightarrow B^2 = (1.02 \times 10^{-8}) \times 2\ \mu_0$

Also, $\frac{1}{\sqrt{\mu_0 \varepsilon_0}} = C \Rightarrow \mu_0 = \frac{1}{C^2 \varepsilon_0}$

$$\Rightarrow B^2 = (1.02 \times 10^{-8})\ 2 \times \frac{4\pi \times 9 \times 10^9}{9 \times 10^{16}}$$

$\Rightarrow B \simeq 160$ nT

27. (*b*) Magnetic energy density $\frac{B^2}{2\mu_0}$

$$\frac{\text{Displacement} \times \text{force}}{(\text{Displacement})^3} = \frac{L.MLT^{-2}}{L^3} = (ML^{-1}T^{-2})$$

28. (*b*) Based on theory.

29. (*a*) Fact based

30. [2] $I = \frac{1}{2}c\varepsilon_0 E_0^2 \Rightarrow \frac{8}{4\pi(10)^2} \times \frac{1}{2} = \frac{1}{4} \times c \times \frac{1}{c^2\mu_0} \times E_0^2$

$$\Rightarrow E_0 = \frac{2}{10}\sqrt{\frac{\mu_0 c}{\pi}} \Rightarrow x = 2$$